Evan Villeneuve

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SUMMARY

Software Engineer and recent Computer Science graduate with hands-on experience building and optimizing fullstack systems. Skilled in designing scalable architectures, ensuring data integrity, and improving system performance and reliability.

EDUCATION

University of Toronto **Sep 2020 - Apr 2025**

Honour's Bachelor of Science, Computer Science, Minor in Mathematics

WORK EXPERIENCE

Aversan **May 2023 - Sep 2023**

Software Engineering Intern

- Created a Python program to automatically extract and insert test data from Excel spreadsheets and test stations into a PostgreSQL database, enabling efficient data analysis.
- Designed the program and system to support concurrent operations, maintaining data integrity and consistency through locking mechanisms and transaction management while improving throughput by 38% compared to the previous system.
- Developed a Python application to parse .pdf and .docx files for comments, troubleshooting document-processing issues and improving company response time by up to 50% for external inquiries and internal reviews.
- Developed and tested embedded C firmware for CAN Bus communication protocols, using Python unit testing to validate reliability and performance while collaborating with cross-functional teams to resolve technical issues and ensure quality.

DataAnnotation **Apr 2025 - Present**

Freelance Software Data Analyst / Data Annotator

- Engineered challenging prompts and expansive repositories in Python, C, and Java for AI coding agents, then reviewed and corrected outputs to identify and resolve technical issues.
- Designed RESTful APIs using Flask, FastAPI, and Django to handle requests and responses, integrating middleware for dependable communication between frontend clients and backend services.

PERSONAL PROJECTS

Python Proxy Server

- Built a single-threaded proxy server to handle multiple concurrent local connections using an event-driven design for HTTP pages.
- Implements local caching with user-configurable TTL, reducing redundant upstream requests and lowering average response latency by upwards of 60%.
- Gained experience in network programming and HTTP protocols, deepening understanding of client-server architecture and data flow management.

Mockup Social Media Website

- Gained valuable collaborative experience through working to develop a full-stack application with two other developers following an Agile methodology.
- Personally designed and implemented a backend RESTful user account API using Django and SQLite, supporting CRUD operations for profiles and account settings.
- Implemented token-based authentication and input validation to secure endpoints and prevent invalid updates.
- Integrated the RESTful API with a React frontend to enable responsive profile management and seamless data flow.

Voice & Soundboard Mixer

- Built a real-time audio mixer in Python that combines live mic input with soundboard clips and outputs to a virtual mic for Discord, OBS and games.
- Designed a drift-correcting audio buffer that holds latency near 25 ms with zero dropouts across devices on separate clocks.
- Integrated RNNoise neural network noise suppression through a C library, applied only to the voice, with a limiter on the final mix to prevent clipping.

SKILLS

- **Programming Languages**: Python, C, C#, Java, JavaScript
- **Web, APIs, & Databases**: HTTP, REST, SQL, Redis, Postgres, AWS, HTML
- **Machine Learning & AI**: GenAI, PyTorch, Tensorflow, Prompt Engineering, Neural Networks, Data Analysis
- **Management Skills**: Technical Project Management, Technical Program Management

REFERENCES

Suraj Manesh, Aversan Software Team Lead - smanesh@aversan.com
Sean Swanepoel, Programmer - StorageHive - (647) 331-4729